

EDC
Assignment #1

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Section :

BEE - 3A1

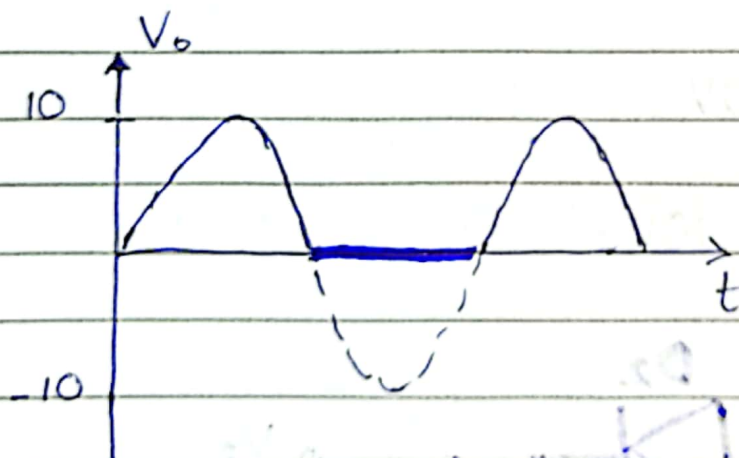
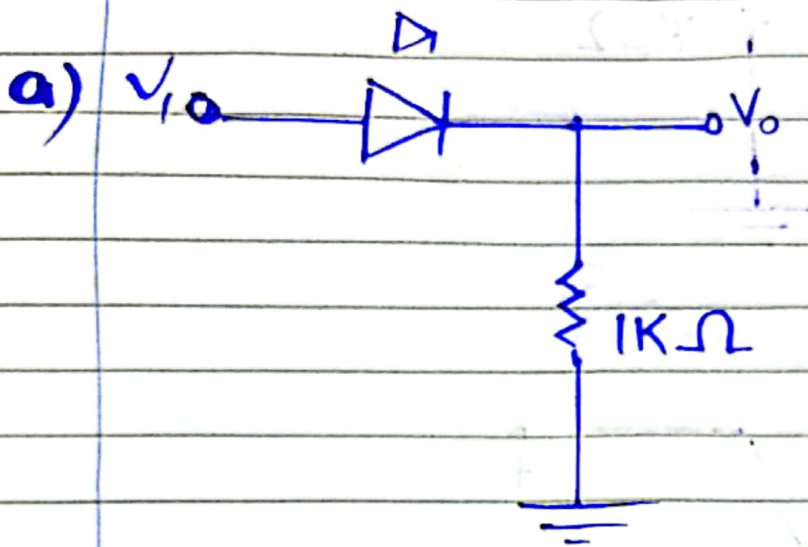
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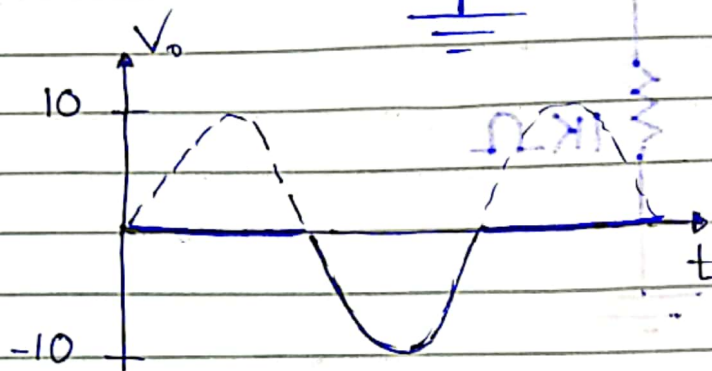
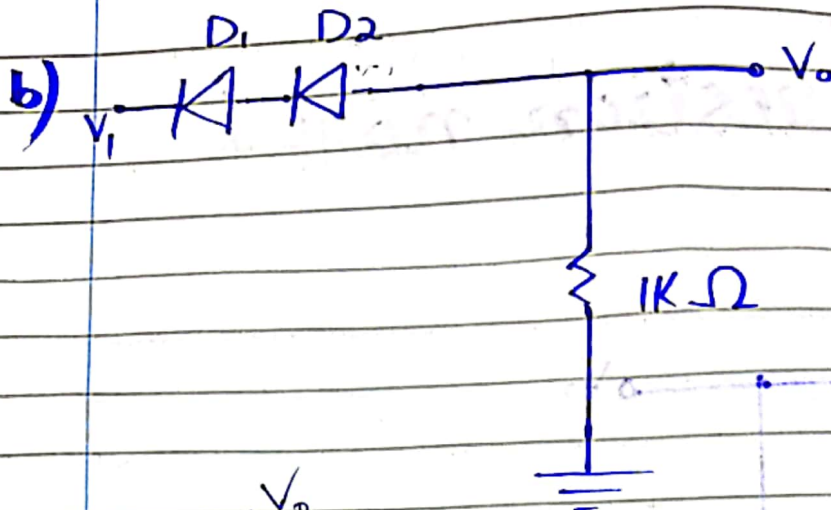
Mam Tarnania

Question no #1



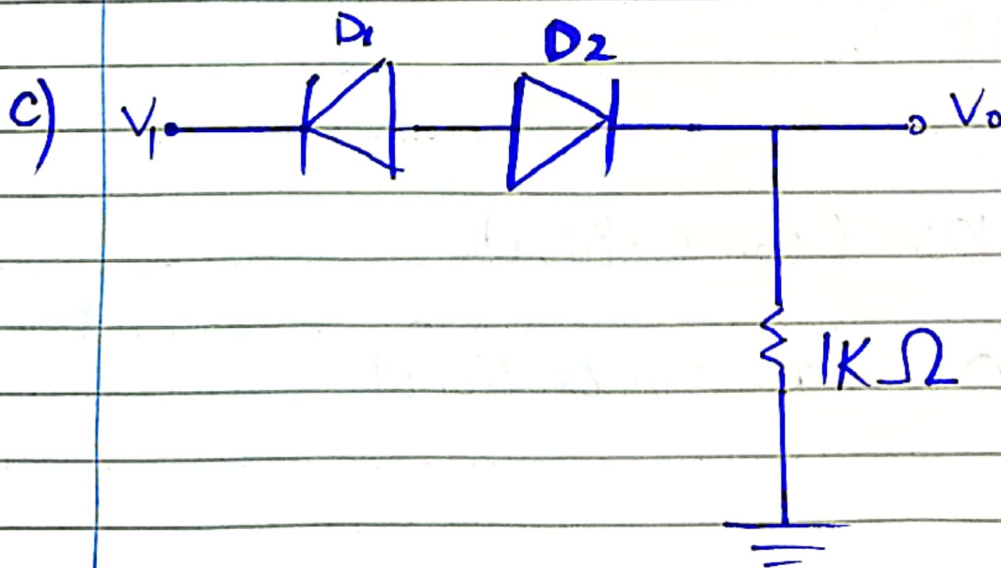
$$V_{p+} = 10V \text{ (conduction)}$$

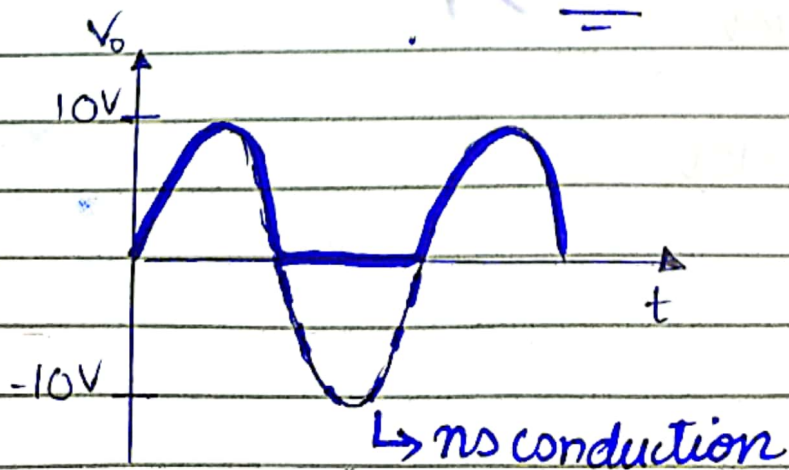
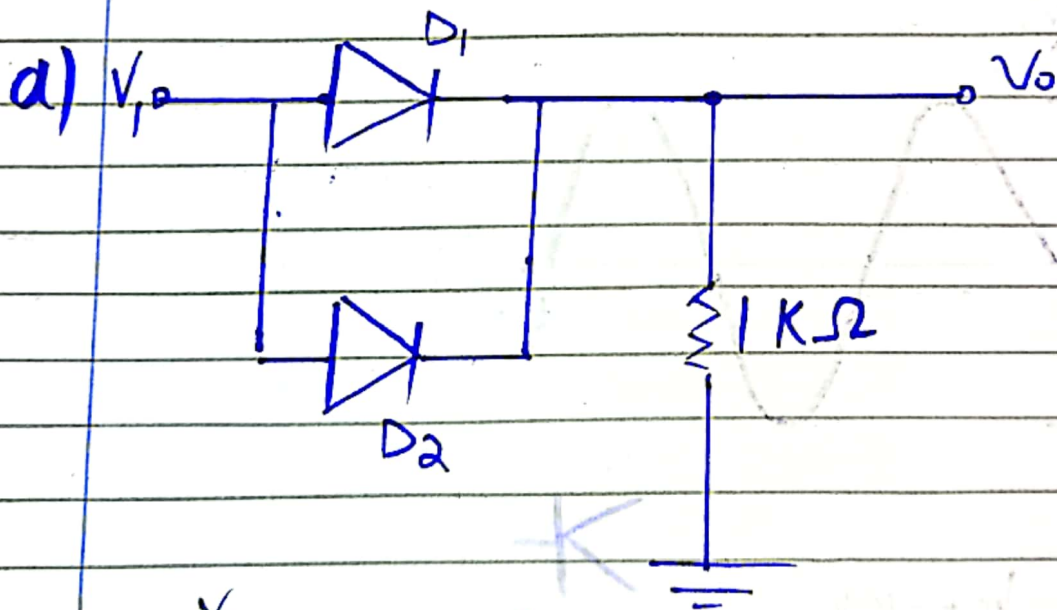
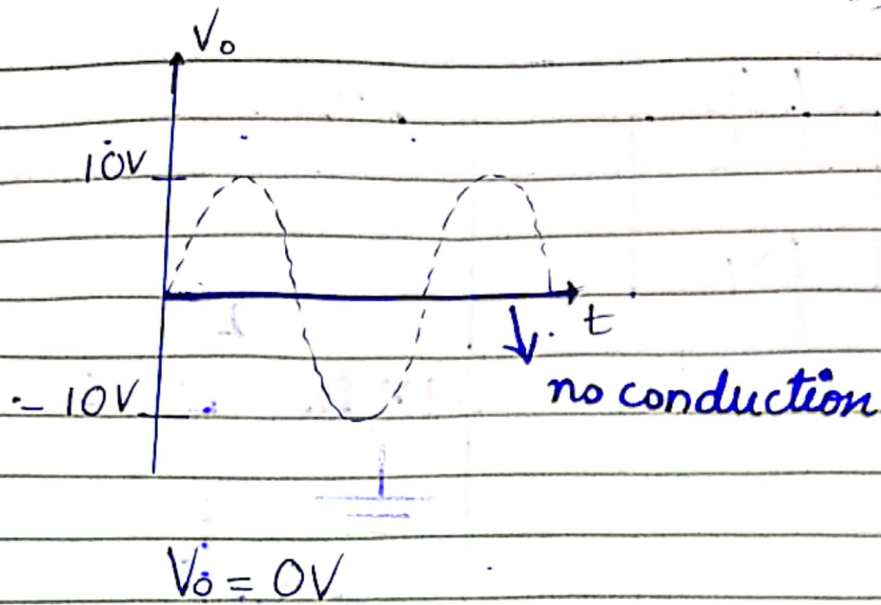
$$V_{p-} = 0V \text{ (no conduction)}$$



$$V_{p+} = 0V$$

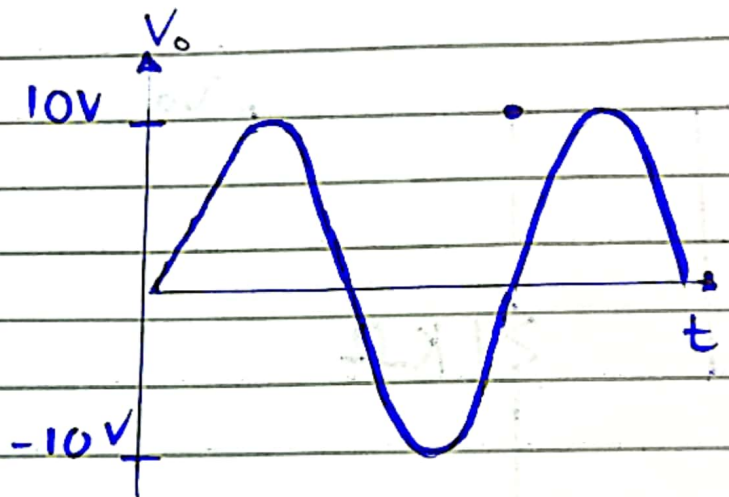
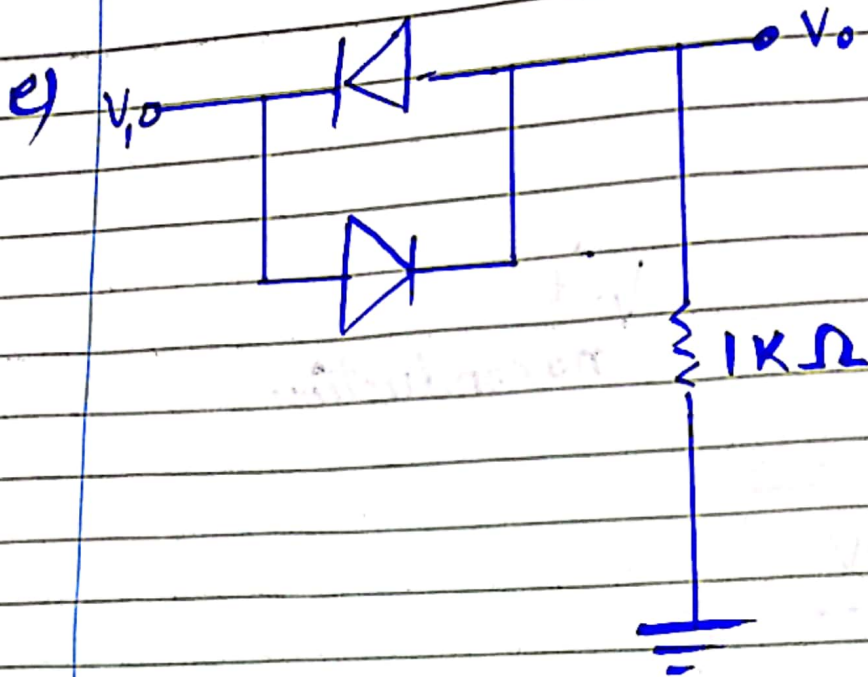
$$V_{p-} = -10V$$





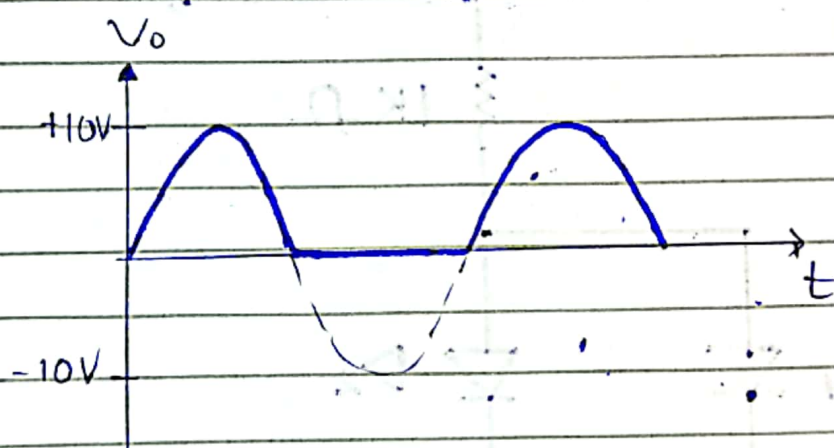
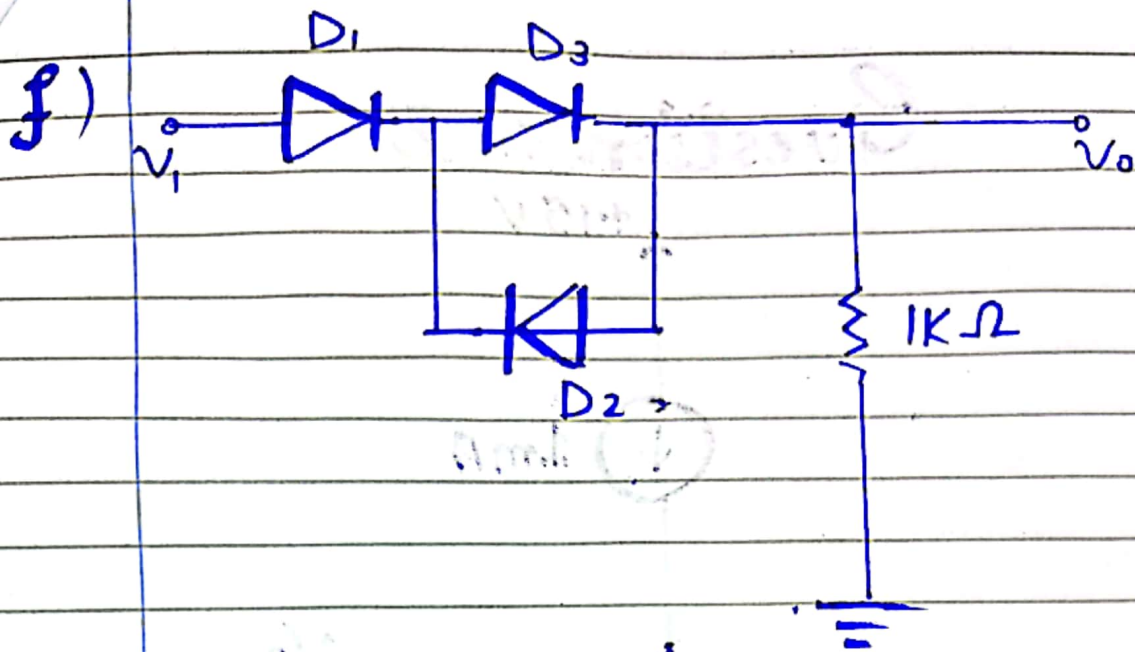
$$V_{p+} = 10V$$

$$V_{p-} = 0V$$



$$V_{p+} = 10V$$

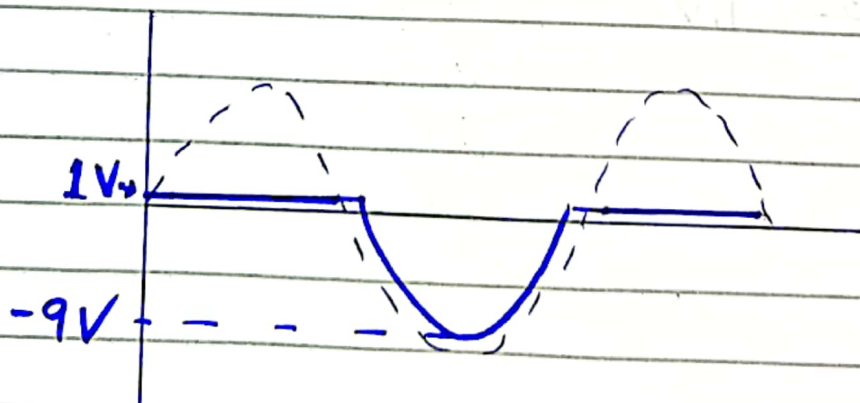
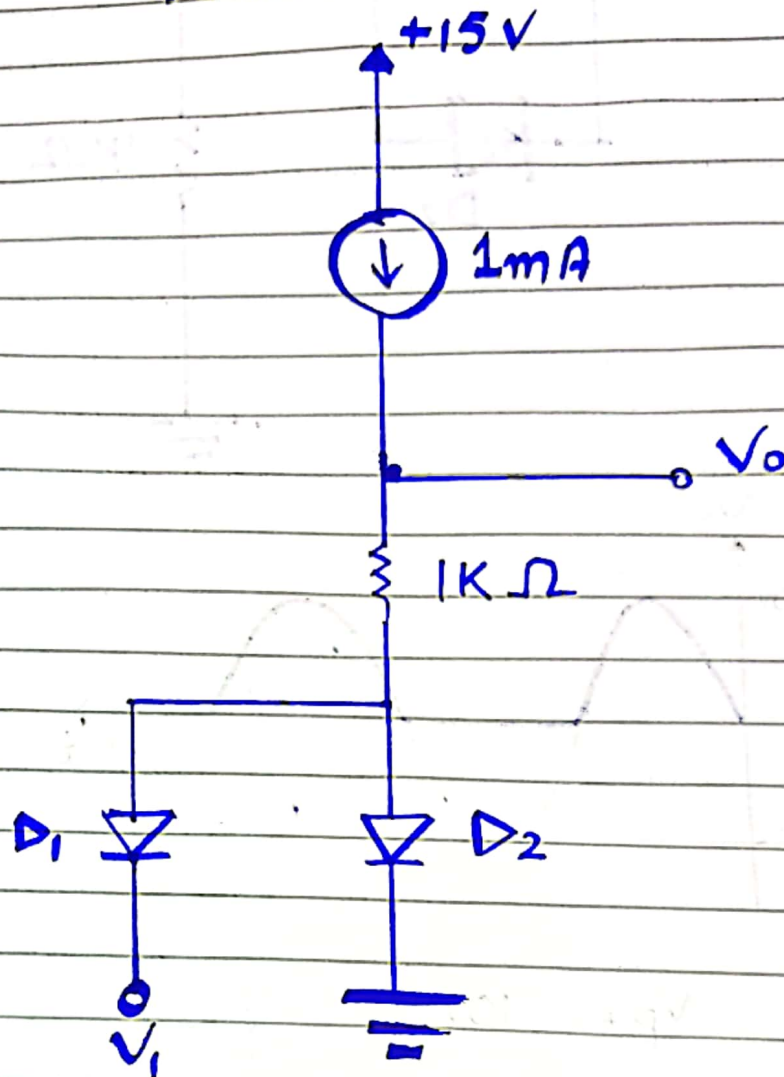
$$V_{p-} = -10V$$



$$V_{p+} = 10V$$

$$V_{p-} = 0V$$

Question #1 2



⇒ When $V_1 > 0V$, means in positive half cycle, D_1 is reverse biased and D_2 is forward biased

So,

$$\boxed{V_o = 1V} \text{ (+ve half cycle)}$$

and

When $V_1 < 0V$, means in negative half cycle, D_1 is conducting (forward biased) and D_2 is not conducting (reverse biased)

So,

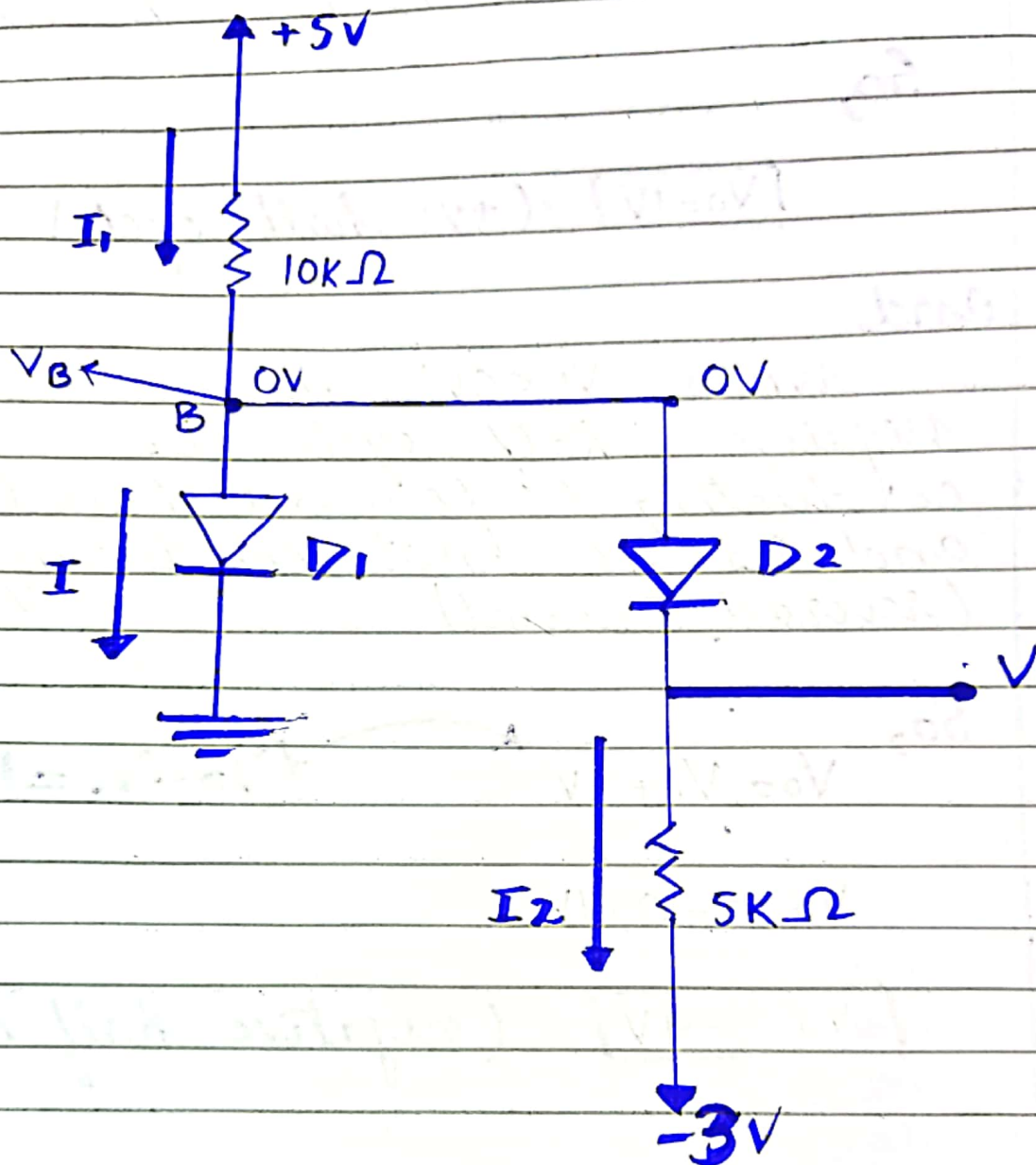
$$V_o = V_1 + 1V$$

$$V_o = -10 + 1V$$

$$(V_o - V_1 = 1V)$$

$$\boxed{V_o = -9V} \text{ (negative half cycle)}$$

Question # 3



Solution :-

Assume D_1 and D_2 F.B

$$V_B = 0V$$

$$\Rightarrow V = 0V$$

finding currents :-

$$I_1 = \frac{5 - 0}{10 \times 10^3}$$

$$I_1 = 5 \times 10^{-4} A$$

Similarly;

$$I_2 = \frac{0 - (-3V)}{5 \times 10^3}$$

$$I_2 = 6 \times 10^{-4} A$$

Applying KCL at node B:-

$$-I_1 + I + I_2 = 0$$

$$I = I_1 - I_2$$

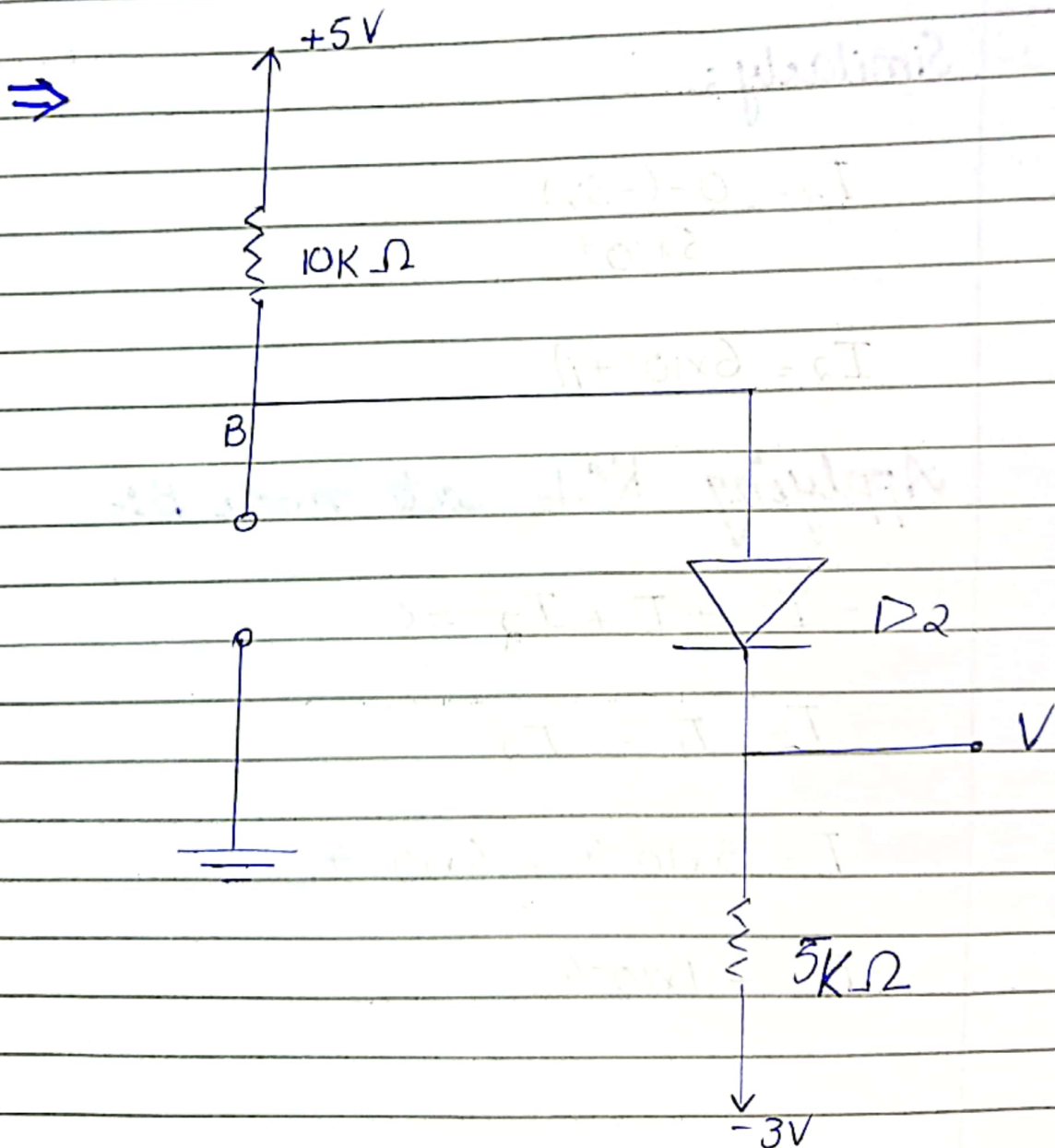
$$I = 5 \times 10^{-4} - 6 \times 10^{-4}$$

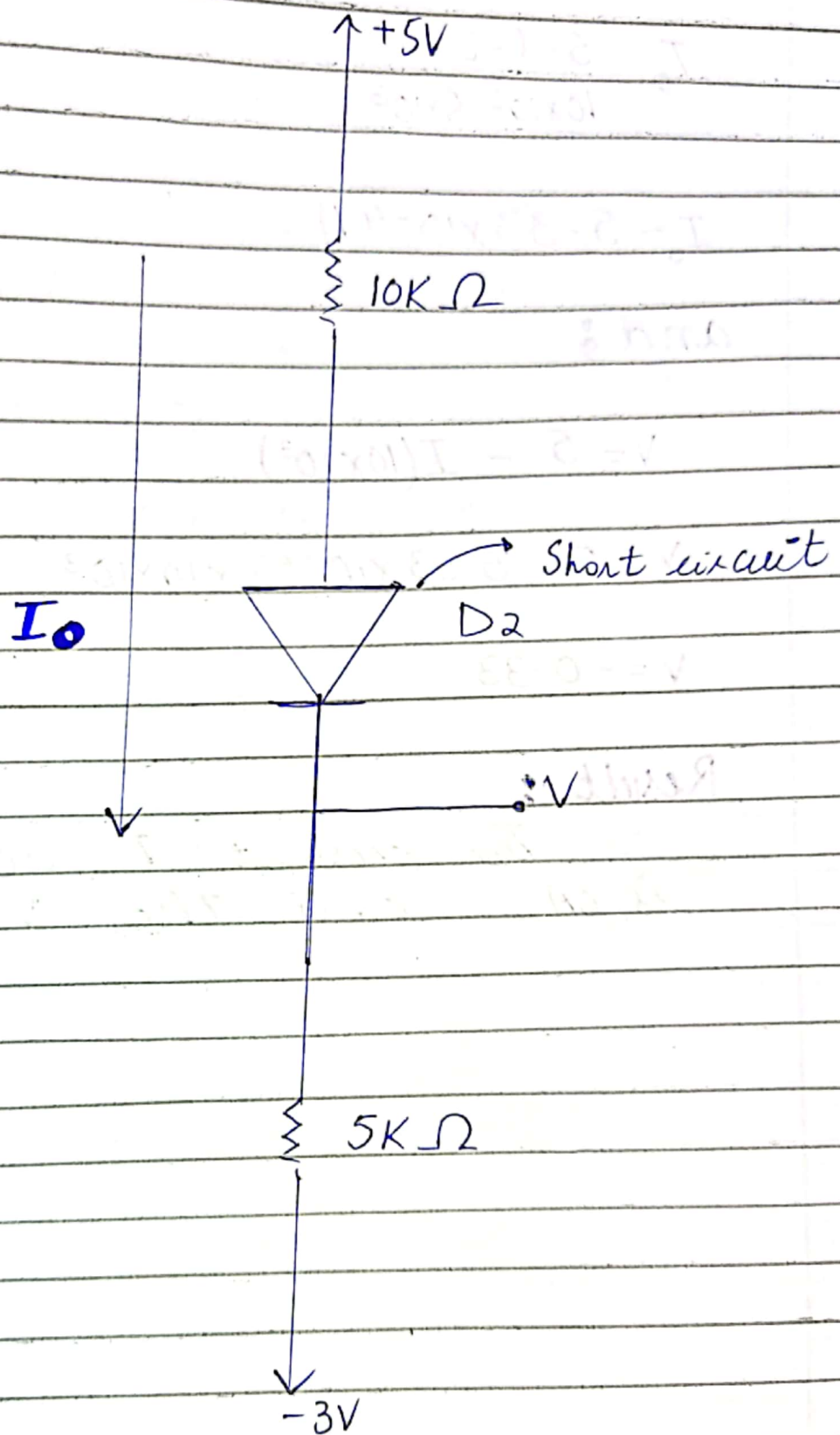
$$I = -1 \times 10^{-4}$$

so, as $I = -1 \times 10^{-4}$, it means that our assumption is wrong and we will have another assumption

Assume;

D_1 reverse bias and D_2 forward bias





$$I_0 = \frac{5 - (-3)}{10 \times 10^3 + 5 \times 10^3}$$

$$I_0 = 5.33 \times 10^{-4} \text{ A}$$

and;

$$V = 5 - I(10 \times 10^3)$$

$$V = 5 - 5.33 \times 10^{-4} \times 10 \times 10^3$$

$$V = -0.33$$

Result :

The current I across D_1 diode is 0 A - and the $V = -0.33 \text{ V}$

Question #4

A particular diode is found to conduct -----?

Solution :-

Using the formula :-

$$i_D = I_S e^{\frac{V_D}{nVT}}$$

Where;

$$i_D = 1 \text{ mA}, \quad V_D = 0.7 \text{ V}, \quad n = 1$$

finding I_S ;

$\Rightarrow$

$$1 \times 10^{-3} = I_S e^{\frac{0.7}{1 \times 25 \times 10^{-3}}}$$

$$\Rightarrow I_S = 6.91 \times 10^{-16}$$

Now;

1) i_d when $V_d = 0.71V$

$$i_d = I_s e^{\frac{V_d}{nV_T}}$$

$$i_d = 6.91 \times 10^{-16} \times e^{\frac{0.71}{1 \times 25 \times 10^{-3}}}$$

$$i_d = 1.49 \times 10^{-3} A$$

2) i_d when $V_d = 0.8V$

$$i_d = I_s e^{\frac{V_d}{nV_T}}$$

$$i_d = 6.91 \times 10^{-16} \times e^{\frac{0.8}{1 \times 25 \times 10^{-3}}}$$

$$i_d = 0.0545 A$$

3) i_d when $V_d = 0.69$

$$i_d = I_s e^{\frac{V_d}{nV_T}}$$

$$i_d = 6.91 \times 10^{-16} \times e^{\frac{0.69}{1 \times 25 \times 10^{-3}}}$$

$$i_d = 6.69 \times 10^{-4}$$

*) i_d when $V_d = 0.6$

$$i_d = I_s e^{\frac{V_d}{n V_T}}$$

$$i_d = 6.91 \times 10^{-16} \times e^{\frac{0.6}{1 \times 25 \times 10^{-3}}}$$

$$i_d = 1.83 \times 10^{-5} \text{ A}$$

Now;

finding change in voltage
when current i_d increases
by factor of 10

$$\Delta V = V_2 - V_1 = 2.3 n V_T \log_{10} \left(\frac{I_2}{I_1} \right)$$

where;

$$I_2 = 10 \times 1 \times 10^{-3}$$

$$I_1 = 1 \times 10^{-3}$$

so;

$$\Delta V = 2.3 \times 1 \times 25 \times 10^{-3} \times \log_{10} \left(\frac{10 \times 1 \times 10^{-3}}{1 \times 10^{-3}} \right)$$

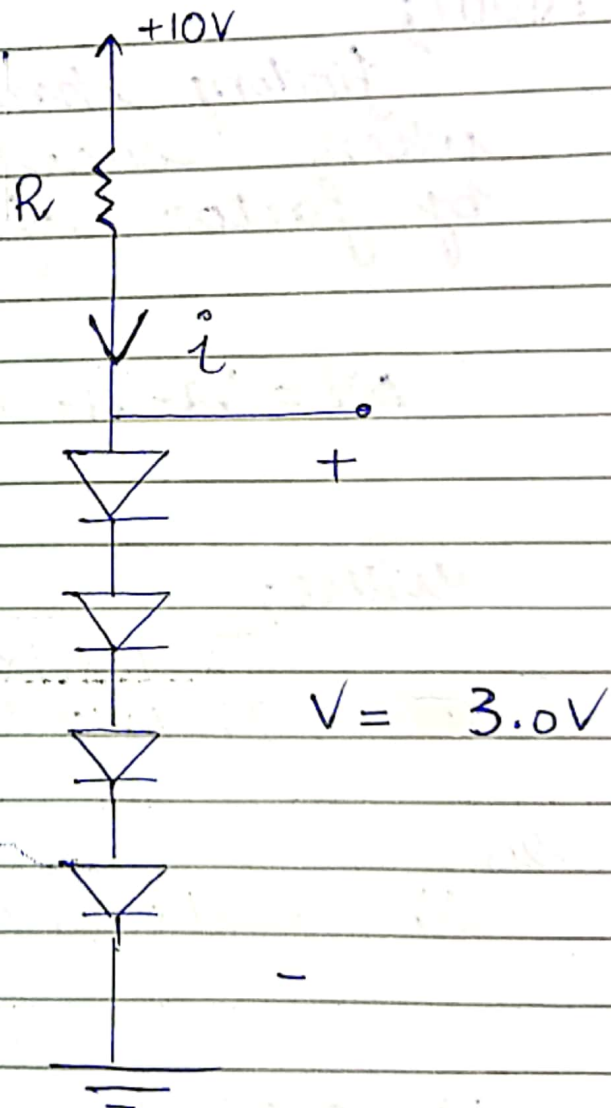
$$\Delta V = 0.0575 \text{ V}$$

Question # 5

Assuming the availability

Solution :-

Circuit :-



Step 1:-
first of one

Step 1:-

first we will find the resistance of one diode, which will be

$$R_d = \frac{V_d}{i_d}$$

$$r_d = \frac{0.7}{1 \times 10^{-3}}$$

$$r_d = 700 \Omega \text{ or } 0.7 \text{ K}\Omega$$

→ As there are 4 diodes connected in series, so total resistance will be;

$$\Rightarrow 4 \times r_d = 2.8 \times 10^3 \Omega$$

$$r_{d \text{ total}} = 2.8 \times 10^3 \Omega$$

Step 2:-

finding current " i "

$$i = \frac{V}{R} = \frac{V_0}{R_{d \text{ total}}}$$

$$i = \frac{3}{2.8 \times 10^3}$$

$$i = 1.07 \times 10^{-3}$$

Step 3:-
finding "R" :-

$$R = \frac{10 - V_0}{i}$$

$$R = \frac{10 - 3}{1.07 \times 10^{-3}}$$

$$R = 6.5K \Omega$$

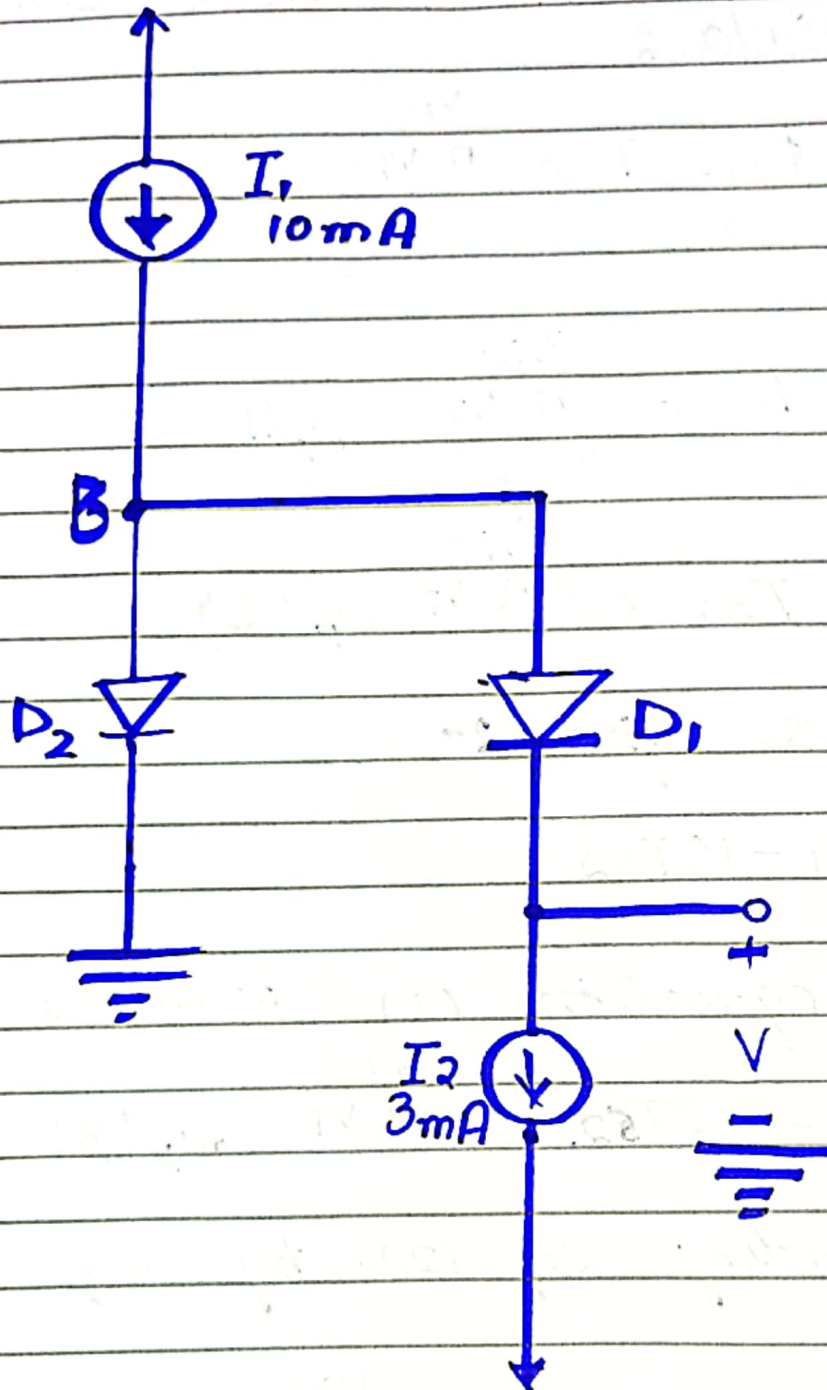
Or

$$R = \frac{10}{i}$$

$$R = \frac{10}{1.07 \times 10^{-3}}$$

$$R = 9.3K \Omega$$

Question # 6



Solution :-

→ In general we will use the formula;

$$I_D = I_s e^{\frac{V_D}{n V_T}}$$

so;

$$I_{D1} = I_{s1} e^{\frac{V_{D1}}{n V_T}} \rightarrow (1)$$

and;

$$I_{D2} = I_{s2} e^{\frac{V_{D2}}{n V_T}} \rightarrow (2)$$

Condition given :-

$$I_{s1} = 10 I_{s2}$$

so equation (1) becomes;

$$I_{D1} = 10 I_{s2} e^{\frac{V_{D1}}{n V_T}} \rightarrow (3)$$

Dividing eq (2) by eq (3)

so;

$$\frac{I_{D2}}{I_{D1}} = \frac{I_{s2} e^{\frac{V_{D2}}{n V_T}}}{10 I_{s2} e^{\frac{V_{D1}}{n V_T}}}$$

$$\frac{I_{D2}}{I_{D1}} = 0.1 e^{\frac{V_{D2} - V_{D1}}{\eta V_T}} \rightarrow (4)$$

As we know;

$$V_o = V_2 - V_1 = \eta V_T \ln \frac{I_2}{I_1}$$

$$V_o = V_2 - V_1 = \eta V_T \ln \frac{10 I_{D2}}{I_{D1}} \rightarrow (5)$$

"Applying KCL at node B:-"

$$\Rightarrow I_1 = I_{D2} + I_{D1}$$

$$I_{D2} = I_1 - I_{D1}$$

where

$$I_1 = 10 \text{ mA}, \quad I_{D1} = 3 \text{ mA}$$

so;

$$I_{D2} = 10 \times 10^{-3} - 3 \times 10^{-3}$$

$$I_{D2} = 7 \text{ mA}$$

finding V

$$V = n V_T \ln \frac{I_2 D_2}{I_1}$$

$$V = 1 \times 25 \times 10^{-3} \times \ln \frac{10 \times 7m}{3 \times 10^{-3}}$$

$$V = 0.0787V$$

Question # 7

Solution :-

As a diode that has a voltage V_D , a resistor R , and a voltage source V are connected in series, the current through the diode can be calculated as given below;

→ When $V_D = 0.7$, the diode current is;

$$I_{D1} = \frac{V - V_D}{R}$$

$$I_{D1} = \frac{V - 0.7V}{R}$$

→ When $V_D = 0.6V$, the diode current is;

$$I_{D2} = \frac{V - V_D}{R}$$

$$I_{D2} = \frac{V - 0.6V}{R}$$

The percent difference in diode current is 1% ;

Where the above statement can be represented mathematically :-

$$I\% \text{ or } 1\% = \frac{|I_{D1} - I_{D2}|}{\left(\frac{I_{D1} + I_{D2}}{2}\right)} \times 100$$

$$1\% = \frac{\left| \left(\frac{V - 0.7}{R} \right) - \left(\frac{V - 0.6}{R} \right) \right|}{\left(\frac{\left(\frac{V - 0.7}{R} \right) + \left(\frac{V - 0.6V}{R} \right)}{2} \right)} \times 100$$

$$\Rightarrow 1 = \frac{0.1}{V - 0.65V} \times 100$$

$$V - 0.65V = 10.0V$$

$$V = 10.0V + 0.65V$$

$$\boxed{V = 10.65V}$$

Then value of V is $10.65V$

As;

$$R = 1K\Omega ; \quad V = 3V$$

So the diodes currents can be calculated as;

$$\text{as;} \quad I_{D1} = \frac{V - 0.7}{1K}$$

$$I_{D1} = \frac{3 - 0.7}{1 \times 10^3}$$

$$I_{D1} = 2.3 \times 10^{-3}A$$

$$I_{D1} = 2.3mA$$

And;
Similarly:-

$$I_{D2} = \frac{V - 0.6}{1 \times 10^3}$$

$$I_{D2} = \frac{3 - 0.6}{1 \times 10^3}$$

$$I_{D2} = 2.4 \text{ mA}$$

Diodes - Current :-

$$I_{D1} = 2.3 \text{ mA} \quad ; \quad I_{D2} = 2.4 \text{ mA}$$

Percentage difference in the two diodes current :-

$$I \% = \frac{|I_{D1} - I_{D2}|}{\left(\frac{I_{D1} + I_{D2}}{2}\right)} \times 100$$

$$I \% = \frac{|2.3 \times 10^{-3} - 2.4 \times 10^{-3}|}{\left(\frac{2.3 \times 10^{-3} + 2.4 \times 10^{-3}}{2}\right)} \times 100$$

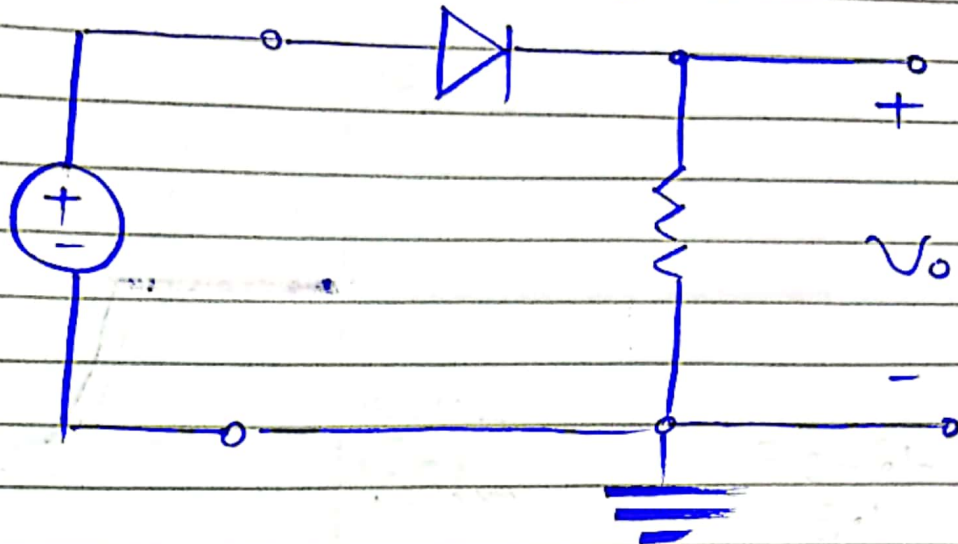
$$I \% = 4.26 \%$$

Results :-

The diodes currents are found to be $I_{D1} = 2.3 \text{ mA}$, $I_{D2} = 2.4 \text{ mA}$, and the percentage difference

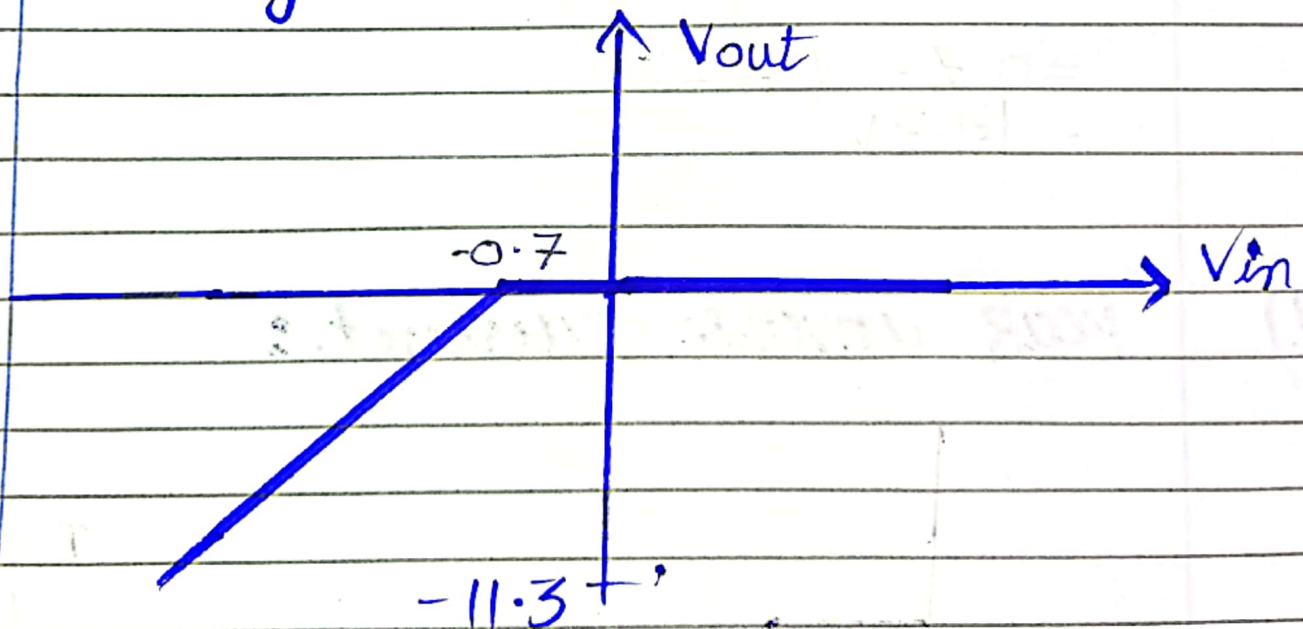
in currents is 4.26%

Question # 8

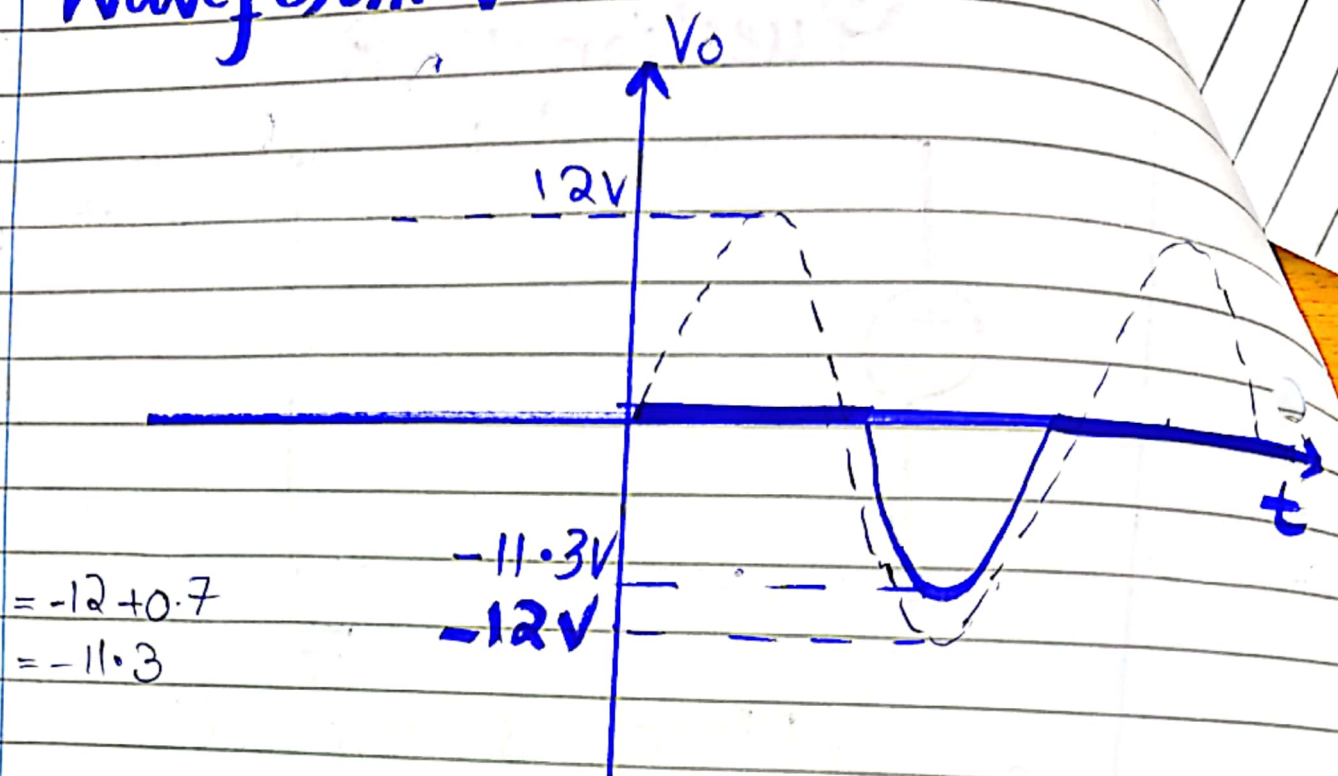


(Diode is reversed)
Solution :-

a) Transfer characteristics :-



b) Waveform V_o :-

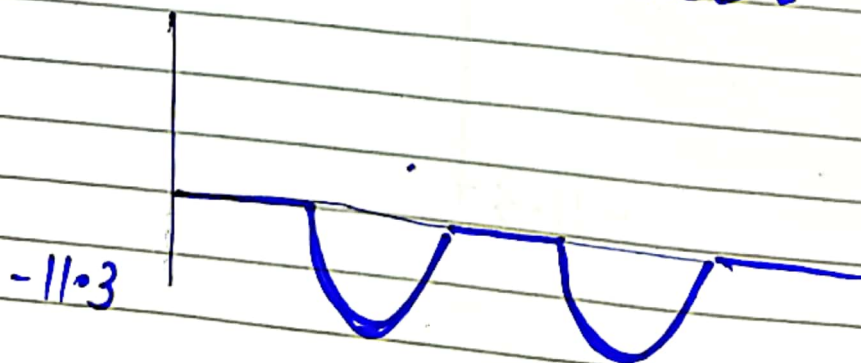


c) Peak Inverse Voltage (PIV) :-

$$= 0.7 - (-12)$$

$$= 12.7V$$

d) Peak inverse current :



$$I = 11.3$$

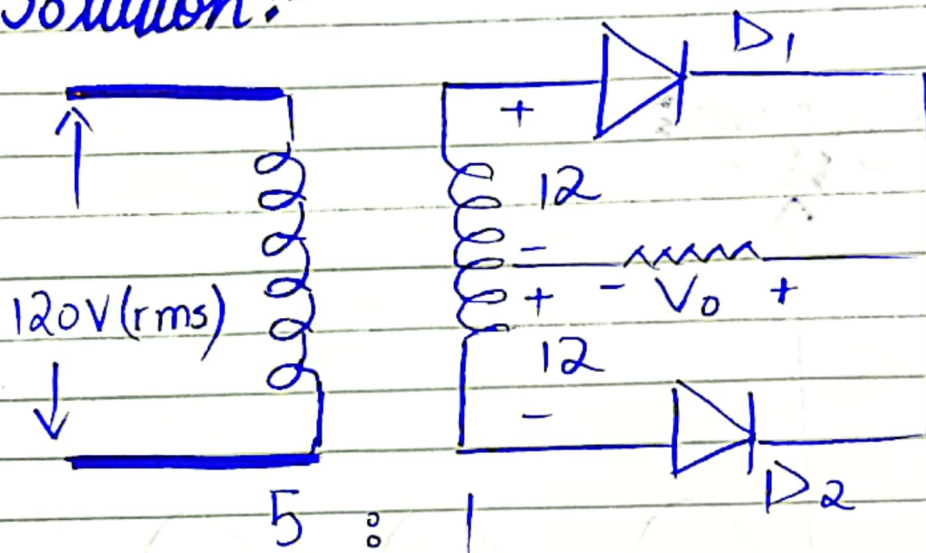
$$1.5 \times 10^{-3}$$

$$I = 7.5mA$$

Question # 9

A full-wave rectifier circuit with $1\text{K}-\Omega$ load operates from a 120V (rms) — — —

Solution:-



In positive half cycle D_1 conducts and in negative half cycle D_2 conducts.

as we know;

$$\frac{V_1}{V_2} = \frac{N_2}{N_1}$$

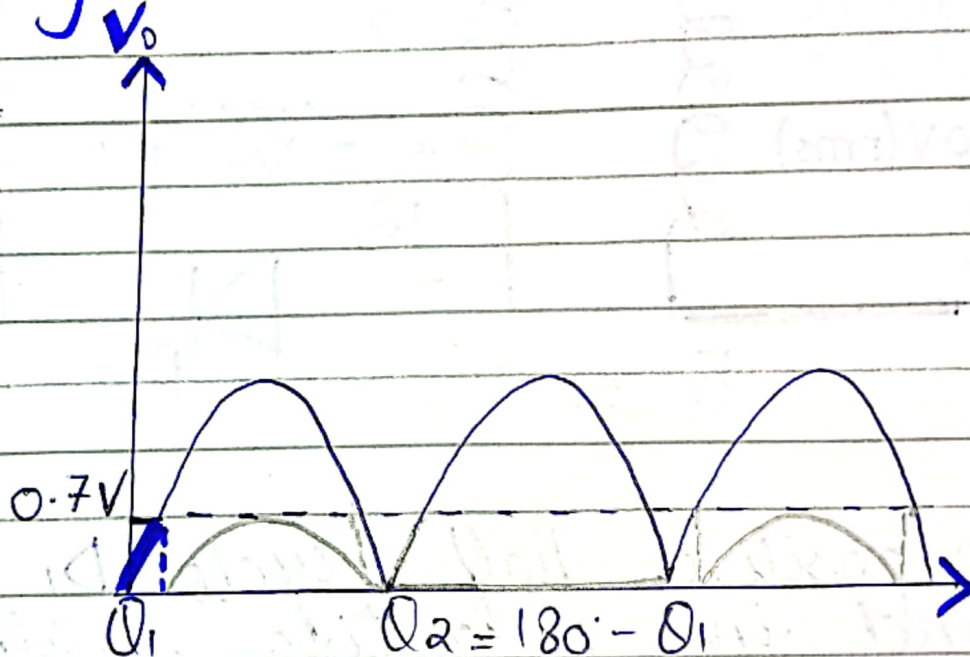
$$\therefore V_1 = 120V$$

$$\frac{120}{V_2} = \frac{N_2}{N_1}$$

$$\frac{120}{V_2} = \frac{5}{1}$$

$$V_2 = 24V$$

Waveform:-



forward bias:-

$$12\sqrt{2} \sin Q = 0.7$$

$$Q = \sin^{-1} \left(\frac{0.7}{12\sqrt{2}} \right)$$

$$\boxed{\theta_1 = 2.36^\circ}$$

*) peak value of rectified output :-

$$= 12\sqrt{2} - 0.7$$

$$= 16.27V$$

*) fraction of diode conduct :-

$$= \frac{180^\circ - 2\theta_1}{180^\circ}$$

$$= \frac{180^\circ - 2 \times 2.36^\circ}{180^\circ}$$

$$= 0.973$$

*) Average of peak voltage :-

$$V_{o\text{avg}} = \frac{1}{\pi} \int_{\theta_1}^{\pi - \theta_1} 12\sqrt{2} \sin \theta d\theta$$

$$V_{o \text{ avg}} = \frac{1}{\pi} \left[12\sqrt{2} \left[-\cos 0 \right]_{\theta_1}^{\pi - \theta_1} \right]$$

$$V_{o \text{ avg}} = \frac{-12\sqrt{2}}{\pi} \left[\cos(\theta_1) - \cos(\pi - \theta_1) \right]$$

$$V_{o \text{ avg}} = \frac{-12\sqrt{2}}{\pi} \left[\cos(2.36^\circ) - \cos(180^\circ - 2.36^\circ) \right]$$

$$V_{o \text{ avg}} = \frac{-12\sqrt{2}}{\pi} \left[0.999 + 0.999 \right]$$

$$V_{o \text{ avg}} = -10.79 \text{ V}$$

*) **Average Current in the Load:-**

$$I_{o \text{ avg}} = \frac{V_{o \text{ avg}}}{R}$$

$$I_{o \text{ avg}} = \frac{10.79}{1 \times 10^3}$$

$$I_{o \text{ avg}} = 0.01079 \text{ A}$$

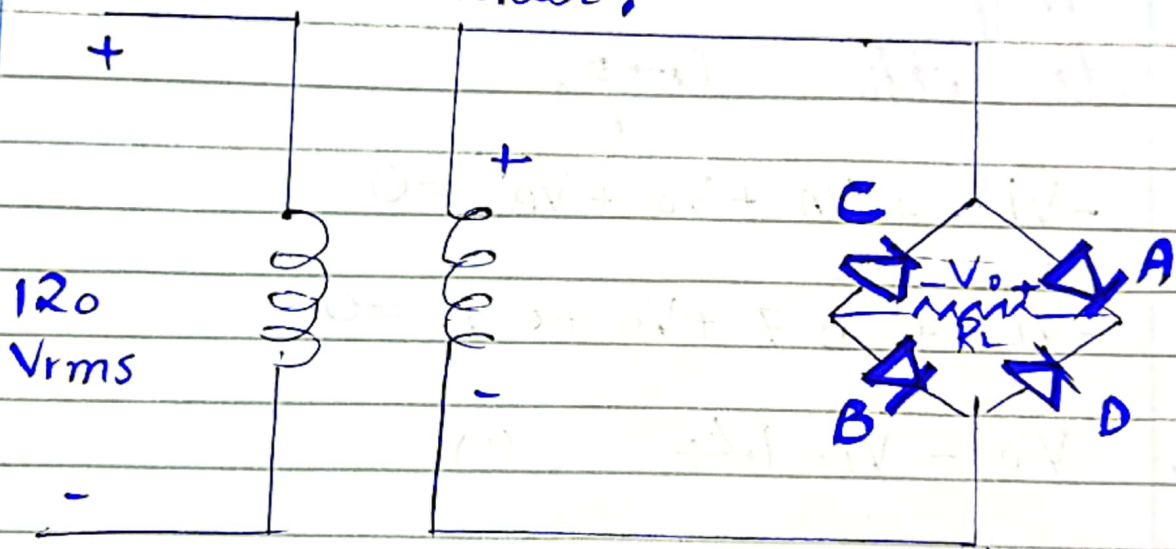
Question # 10

It is required to design full wave rectifier circuit ---

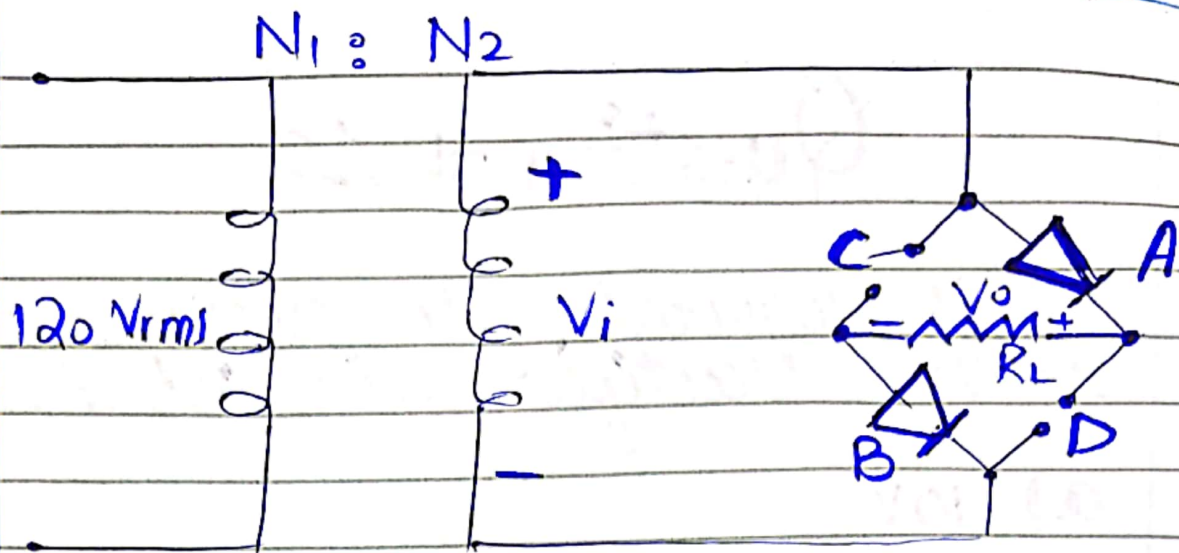
a) 10V

b) 100V

Consider :-



For positive half cycle of the input, the diodes **A** and **B** conduct



Apply the Kirchhoff's voltage to the loop.

$$-V_i + V_A + V_o + V_B = 0$$

$$-V_i + 0.7 + V_o + 0.7 = 0$$

$$\Rightarrow V_o = V_i - 1.4 \rightarrow (1)$$

The Average output voltage is;

$$V_{o \text{ avg}} = \frac{2V_o}{\pi} \rightarrow (2)$$

from eq (1)

$$V_o = V_i - 1.4 \text{ (Put in eq (2))}$$

$$V_{o \text{ avg}} = \frac{2(V_i - 1.4)}{\pi}$$

a) for $V_{o \text{ avg}} = 10\text{V}$

$$\Rightarrow \frac{2(V_i - 1.4)}{\pi} = V_{o \text{ avg}}$$

$$\frac{2(V_i - 1.4)}{\pi} = 10$$

$$(V_i - 1.4) = \frac{10\pi}{2}$$

$$V_i = 5\pi + 1.4$$

$$V_i = 17.108 \text{ V}$$

The peak value of ac line voltage :-

$$V_{ac} = 120\sqrt{2} \text{ V}$$

Voltage turns ratio :-

$$\frac{N_1}{N_2} = \frac{V_{ac}}{V_i}$$

$$\frac{N_1}{N_2} = \frac{120\sqrt{2}}{17.108}$$

$$\frac{N_1}{N_2} = 9.919$$

$$\frac{N_1}{N_2} \approx 10$$

b) for $V_{oavg} = 100V$

$$V_{oavg} = \frac{2(V_i - 1.4)}{\pi}$$

$$100 = \frac{2(V_i - 1.4)}{\pi}$$

$$\frac{100\pi}{2} = V_i - 1.4$$

$$50\pi + 1.4 = V_i$$

$$\Rightarrow V_i = 158.47V$$

The peak value of ac line voltage

$$V_{ac} = 120\sqrt{2}V$$

from the voltage turns ratio :-

$$\frac{N_1}{N_2} = \frac{V_{ac}}{V_i}$$

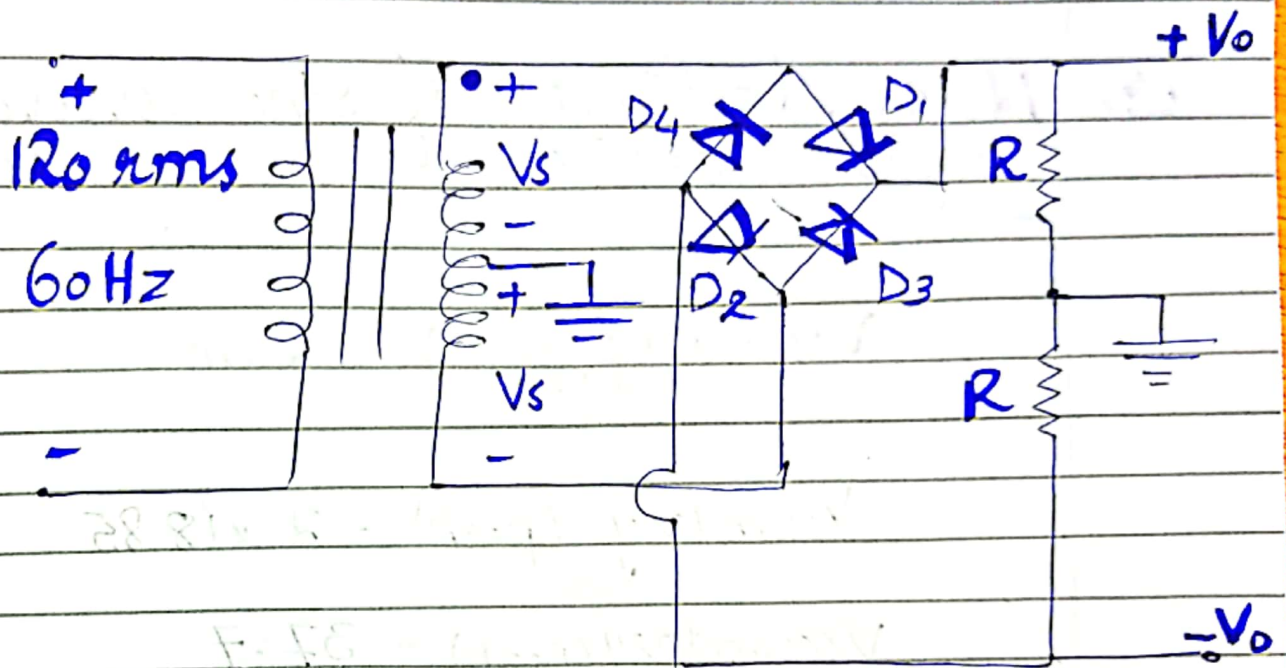
$$\frac{N_1}{N_2} = \frac{120\sqrt{2}}{158.48}$$

$$\frac{N_1}{N_2} = 1.07$$

$$\frac{N_1}{N_2} \approx 1$$

Question # 11

Solution :-



The magnitude of each average output voltage is:

$$= 15V$$

⇒ output is similar to full wave Rectifier

Out put:-

$$V_{avg} = \frac{2V_m}{\pi}$$

$$V_m = \frac{V_{avg} \times \pi}{2}$$

$$V_m = 23.56 \text{ V}$$

⇒ Now, Amplitude of sine wave across entire secondary winding - ϕ

$$V_{secondary}(\text{peak}) = 2V_m$$

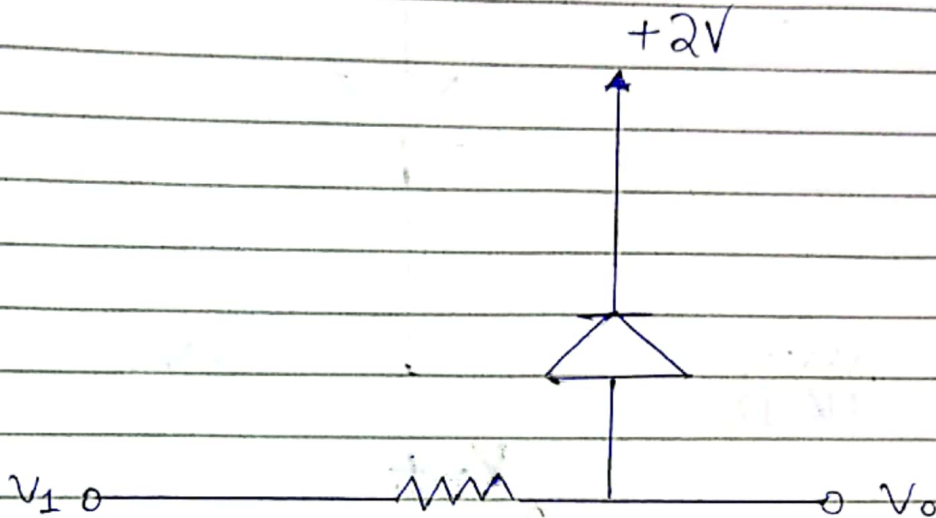
$$V_{secondary}(\text{peak}) \approx 2 \times 23.56$$

$$V_{secondary}(\text{peak}) = 47.12$$

Question # 12

Solutions :-

a)

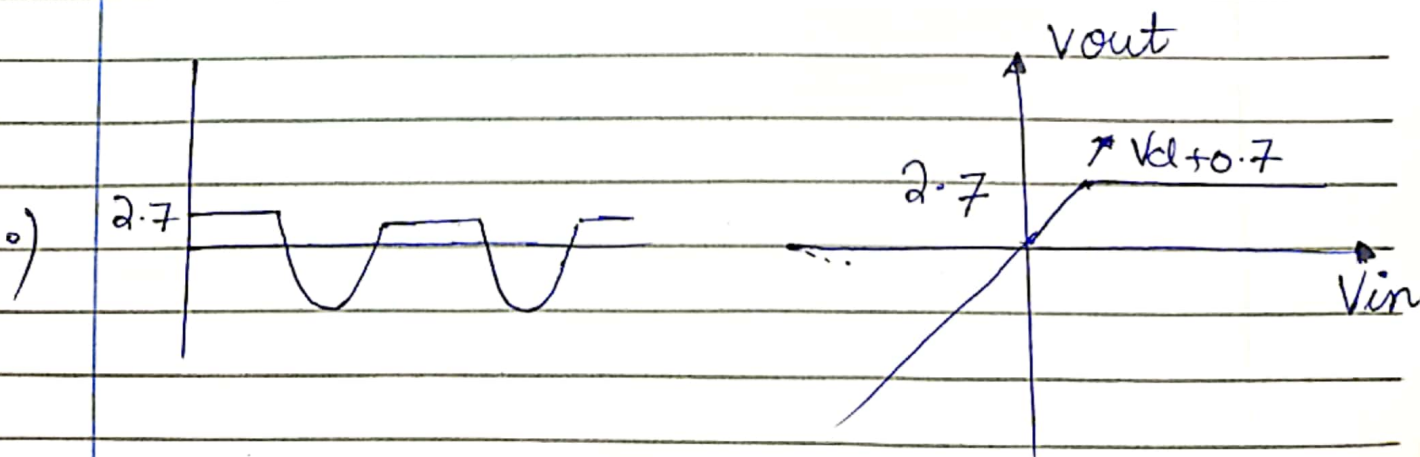


$$V_1 = 12 \sin t$$

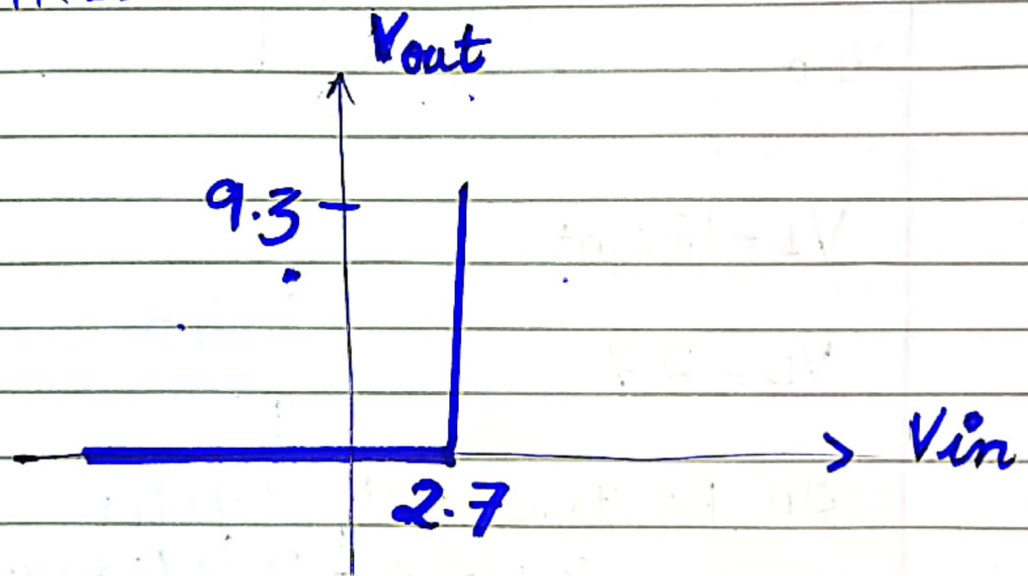
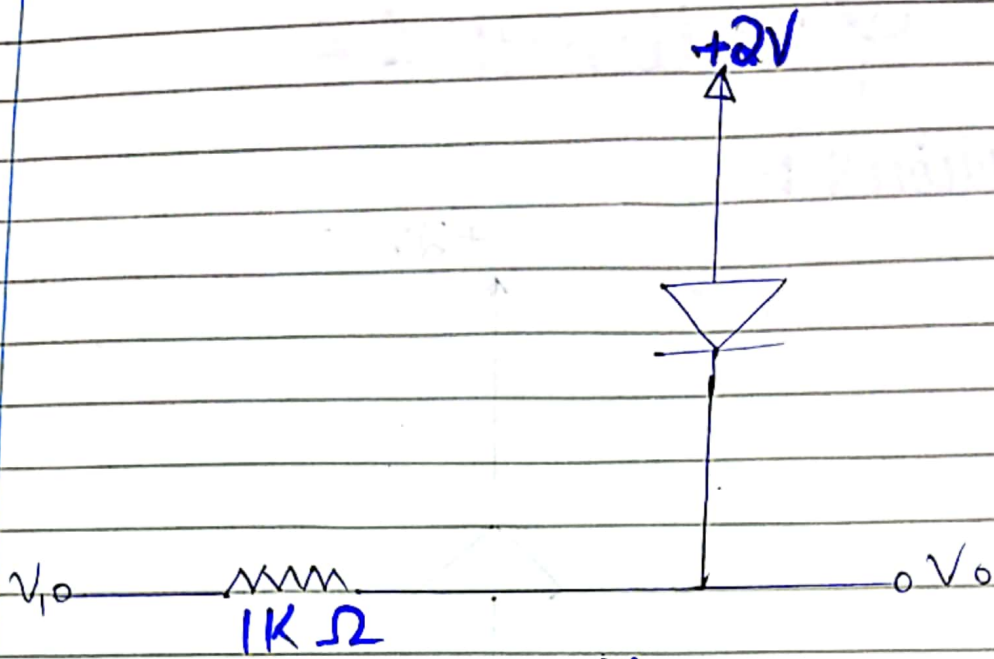
$$V_D = 2.7$$

In positive half cycle:-

$$2 + 0.7 = 2.7 \text{ (conduction)}$$



b)



c)

V_i

$1K\Omega$

V_o



$-2V$

V_{out}

